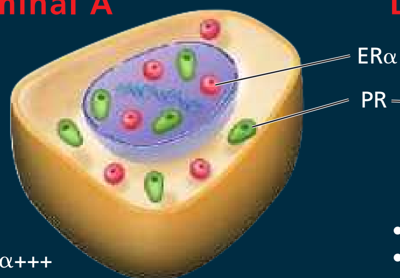
Breast Cancer Subtypes

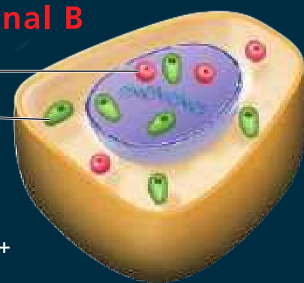
Each breast cancer subtype is characterized by the overexpression (indicated by ++ or +++) or absence (–) of three signal receptors: ERα, PR, and HER2. Breast cancer treatments are becoming more effective because they can be tailored to the specific cancer subtype.

Luminal A



- ERα+++
- PR++
- HER2–
- 40% of breast cancers
- Best prognosis

Luminal B



- ERα++
- PR++
- HER2– (shown here); some HER2++
- Divide rapidly
- 15–20% of breast cancers
- Poorer prognosis than luminal A

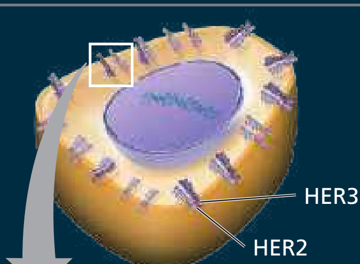
Both luminal subtypes overexpress ERα (luminal A more than luminal B) and PR, and usually lack expression of HER2. Both can be treated with drugs that target ERα and inactivate it, the most well-known drug being tamoxifen. These subtypes can also be treated with drugs that inhibit estrogen synthesis.

Basal-like

- ERα–
- PR–
- HER2–
- 15–20% of breast cancers
- More aggressive; poorer prognosis than other subtypes

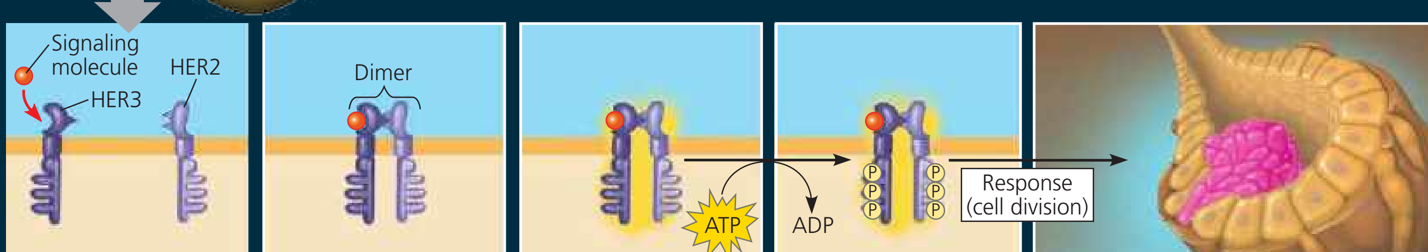
The basal-like subtype is "triple negative"—it usually does not express ERα, PR, or HER2. It often has a mutation in the tumor-suppressor gene *BRCA1* (see Concept 18.5). Treatments that target ERα, PR, or HER2 are not effective, but new treatments are being developed. Currently, patients are treated with cytotoxic chemotherapy, which selectively kills fast-growing cells.

HER2



- ERα– or ERα+
- PR–
- HER2++
- 10–15% of breast cancers
- Poorer prognosis than luminal A subtype

The HER2 subtype overexpresses HER2. Because it does not express PR (and, in many cases, ERα) at normal levels, the cells are unresponsive to therapies that target PR (and, for ERα– cells, ERα). However, patients with the HER2 subtype can be treated with Herceptin, an antibody protein that inactivates HER2 (see Concept 12.3).



1 A signaling molecule (such as a growth factor) binds as a ligand to a single HER3 (a monomer), allowing it to associate with another HER.

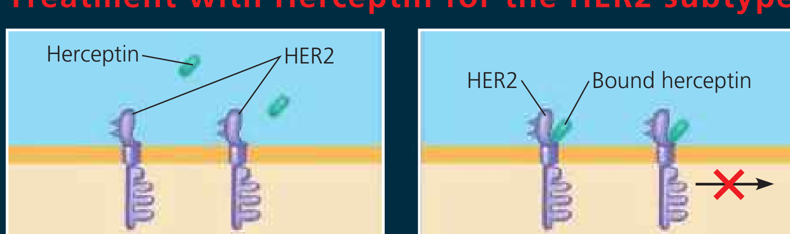
2 Even without a ligand bound, HER2 can associate with another HER. Here, HER3 and HER2 associate closely together, forming a dimer.

3 Formation of a dimer activates each monomer.

4 Each monomer adds phosphate from ATP to the other monomer, triggering a signal transduction pathway.

5 The signal is transduced through the cell, which leads to a cellular response—in this case, turning on genes that trigger cell division. HER2 cells have up to 100 times as many HER2 receptors as normal cells, so they undergo uncontrolled cell division.

Treatment with Herceptin for the HER2 subtype



1 Patients with the HER2 subtype are treated with the drug Herceptin, which can bind to HER2.

2 When bound to HER2, Herceptin blocks signaling and excessive cell division in certain patients.

MAKE CONNECTIONS

When researchers compared gene expression in normal breast cells and cells from breast cancers, they found that the genes showing the most significant differences in expression encoded signal receptors, as shown here. Given what you learned in Chapters 11, 12, and this chapter, explain why this result is not surprising.